Dynamic Performance Evaluation of Sensorless Permanent-Magnet Synchronous Motor Drives With Reduced Current Sensors

Matteo Carpaneto, Paolo Fazio, Mario Marchesoni, *Member, IEEE*, and Gianluca Parodi

***Abstract*—A novel position sensorless permanent-magnet synchronous motor (PMSM) drive that uses only a direct current (dc)-link sensor is presented in this paper. This concept is already known, but several new implementation details are shown in this paper. The proposed drive system is simple and can be implemented using low-cost control hardware. The dynamic performance of the proposed drive is evaluated and compared with the performance of position sensorless and full sensored drives.**

***Index Terms*—Direct current (dc)-link current sensor, low-cost implementation, permanent-magnet synchronous motors (PMSMs), sensorless control.**

I. Introduction

PERMANENT-MAGNET Synchronous Motors (PMSMs) are very common in industrial applications because of their good qualities such as high robustness, compactness, and efficiency.

PMSM performance is also increased by vector control techniques and, recently, tested magnetic materials [1]. Furthermore, the continuous development of microcontrollers and the consequent cost reduction allowed the use of PMSMs in low-cost drives and general-purpose applications.

It is widely known that synchronous motor control needs a rotor position feedback, which is generally provided by an encoder or a resolver. Sensorless technology basically replaces this position transducer with a position estimator. There are several advantages by using this technology, such as reliability increase and cost decrease. Moreover, the sensorless strategy allows motor working in hostile environment and when available space is very small [2]–[31].

Recently, some contributions have appeared in the technical literature, where a great interest in the reconstruction techniques of the three motor-phase currents, using only the single dc-link current sensor, has been expressed. The interests for this research topic are to be referred to the savings associated with reducing the number of current sensors and to the elimination of the potential load imbalances caused by unequal gains of output terminal current sensors [32]–[39].

In low-power applications, control hardware can also affect the drive final cost in great amount. Due to this reason, the sensorless implementation is not the only key approach for conquering this share of the market, but it also has to necessarily be joined to the employment of low-cost digital electronics. In this case, the maximum sampling frequency and, sometimes, the switching frequency of the power semiconductor devices will turn out to be low enough (some kilohertz at the maximum) to allow the control program to execute the requested operations in real time. Such low frequencies can, however, compromise the motor dynamic and static performance and, in general, cannot easily permit the application of the theories developed and applied in the last years to estimate the motor electrical and mechanical quantities.

A novel position sensorless PMSM drive, which uses only one dc-link current sensor, is presented in this work. The proposed drive system is simple and can be implemented using low-cost control hardware, without additional regulators or space vector modulation (SVM) alteration, as reported in [32], [34], and [36], and without observers that imply a greater computational effort, as shown in [36] and [37].

The method proposed in this paper maintains the classical field-oriented control (FOC) characteristics [40] to maximize dynamic performance: this feature is not present in [35], where the only regulation is based on the minimization of phase difference between voltage and current.

Position and current estimators imply a slight deterioration of the dynamic response. This paper provides a specific evaluation of the dynamic response of the proposed sensorless drive with reduced current sensors by comparing it with traditional full sensored and position sensorless drives, also to overcome the absence of dynamic performance evaluation in the aforementioned papers [35]–[37].

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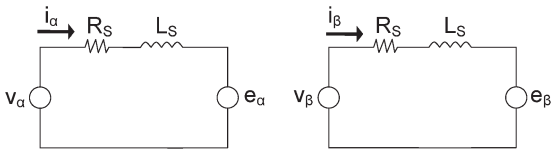


Fig. 1. Synchronous motor-equivalent circuit split into an α–β reference frame.

II. Estimation Algorithms

A. Speed and Position Estimation

Several different ways of estimating rotor position and speed are listed as follows:

1) estimating the back–electromotive force (EMF), using the motor parameters, measured current, and imposed voltage [5]–[7];
2) injecting high-frequency voltage and exploiting the angle, depending on the currents measured [8]–[16], [26]–[28];
3) using a state observer such as an extended Kalman filter or an adaptive observer [17]–[23], [28]–[31].

These strategies do not often provide excellent results in all the operating conditions. For example, back-EMF estimators work reasonably well at medium and high speeds, but their performance deteriorates at low speeds. On the contrary, high-frequency voltage injection is useful in the low-speed region but becomes not practicable when the speed increases. To overcome the specific limitations of these methods, they are often used in combination with each other, as shown, for example, in [24] and [25].

In this paper, a back-EMF method has been chosen to estimate rotor position, although it requires an open-loop start-up strategy. It is not possible to utilize high-frequency injection methods, because high-frequency current could not be provided by the current estimator.

The use of a state observer such as an extended Kalman filter would require great computing power by a control processor, and this condition is in disagreement with a low-cost sensorless strategy.

The back-EMF estimation method is a good way of deducing rotor position, and it is based on a synchronous-machine dynamic-stator model transformed into an α–β reference frame.

The angle estimation runs every SVM cycle time T_C. This cycle time is directly related to the switching frequency f_c. The SVM that is implemented in this paper is a discontinuous pulsewidth modulator (DPWM), as shown in [41] and [42], with the voltage pulses on each phase leg centered within the switching period.

At the beginning of every SVM cycle time, a constant voltage is imposed by the control system, and during this cycle, back-EMF is also considered constant. A dynamic stator model, which is split into an α–β fixed reference frame, gives two resistor–inductor (RL) circuits, as shown in Fig. 1, with constant forcing voltage, from which it is easy to calculate motor-phase currents by solving two differential equations with constant coefficients [43].

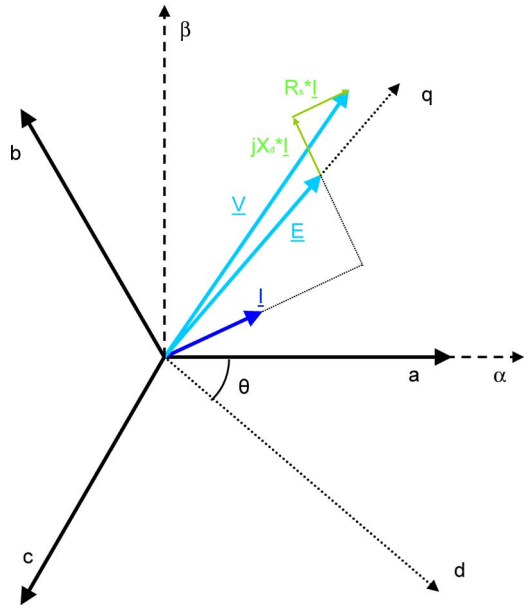


Fig. 2. Synchronous motor vector diagram.

System-state variables exhibit exponential behavior with time constant L_s/R_s [43], i.e.,

$$\begin{cases} i_\alpha(t) = \left(\frac{V_\alpha}{R_s} - \frac{E_\alpha}{R_s} - i_\alpha(0)\right) \cdot \left(1 - e^{-\frac{R_s}{L_s}t}\right) + i_\alpha(0) \\ i_\beta(t) = \left(\frac{V_\beta}{R_s} - \frac{E_\beta}{R_s} - i_\beta(0)\right) \cdot \left(1 - e^{-\frac{R_s}{L_s}t}\right) + i_\beta(0). \end{cases} \tag{1}$$

In these relationships, back-EMFs are the only unknown terms. Equation (2) can easily be deduced, expressed in discrete time and containing the K_T term, which represents the discrete-time exponential [43]

$$\begin{cases} E_\alpha = (V_\alpha - R_s \cdot i_\alpha(n-1)) - R_s \\ \quad \cdot (i_\alpha(n) - i_\alpha(n-1)) \cdot \frac{1}{(1-K_T)} \\ E_\beta = (V_\beta - R_s \cdot i_\beta(n-1)) - R_s \\ \quad \cdot (i_\beta(n) - i_\beta(n-1)) \cdot \frac{1}{(1-K_T)} \end{cases} \tag{2}$$

$$K_T = e^{-\frac{R_s}{L_s}\Delta t}. \tag{3}$$

Back-EMFs, as deduced from (2), are used to find the q-axis angle, as shown in Fig. 2. Adding or subtracting 90° is necessary (depending on whether the rotor speed is negative or positive) to calculate the d-axis angle θ as

$$\theta = \arctan\frac{E_\beta}{E_\alpha} \pm \pi/2. \tag{4}$$

Rotor speed estimation is implemented using a position discrete-time derivative. The discrete period chosen for derivation is equal to ten SVM cycles, i.e.,

$$\omega_j = \frac{\theta(n) - \theta(n-10)}{10 \cdot T_C}. \tag{5}$$

Such a procedure implies the first filtering effect. Moreover, a first-order low-pass discrete filter with cutoff frequency equal to

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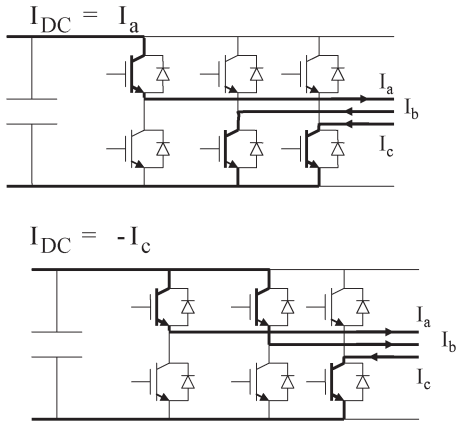


Fig. 3. Examples of two inverter-switch configurations.

100 Hz is implemented. All these steps ensure speed regulator stability.

For clarity, the Clark and Park transformations used in the back-EMF estimator and in other control algorithms are reported as [44]

$$\begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 & -0.5 & -0.5 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \tag{6}$$

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}\begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix}. \tag{7}$$

B. Current Estimation

FOC strategy implementation requires knowledge of the phase currents; phase currents can be reconstructed by dc-link current measurement and by their corresponding switches configurations. Fig. 3 shows how it is possible to associate dc-link current measurement to motor-phase currents when only one high branch is active or two high branches are active. Conversely, if all three high branches are active or inactive at the same time, the dc-link current is zero, as well as the motor voltage.

Each SVM period T_C is composed of the repetitive sequence of the following four switch configurations.

1) No high branches are turned on.
2) Only one high branch is turned on.
3) Two high branches are turned on.
4) All three high branches are turned on.

In each SVM period, there are two different configurations where the association of the dc-link current with two different motor-phase currents is possible (with a positive or negative sign, depending on whether one or two high branches are turned on).

The third current value can be calculated as the negative of the sum of the other two currents (three-wire system). This estimator needs three information from the SVM: the hexagon sectors K_S, where sectors are numbered from 0 to 5, depending on the imposed voltage vector angle δ

$$\frac{\pi}{3}\cdot K_S \le \delta < \frac{\pi}{3}\cdot(K_S+1) \tag{8}$$

and the two space vector permanence times t_1 and t_2, which are functions of the modulation index m

$$m = \frac{\sqrt{V_\alpha^2 + V_\beta^2}}{\frac{\sqrt{3}}{2}\cdot V_{DC}} \tag{9}$$

and calculated with conventional trigonometric functions [42]

$$t_1 = m\cdot\frac{T_C}{2}\cdot\sin\left(\frac{\pi}{3} - \delta + \frac{\pi}{3}K_S\right) \tag{10}$$

$$t_2 = m\cdot\frac{T_C}{2}\cdot\sin\left(\delta - \frac{\pi}{3}K_S\right) \tag{11}$$

$$t_{0,7} = \frac{T_C}{2} - t_1 - t_2 = \frac{t_0}{2} + \frac{t_7}{2}. \tag{12}$$

The current estimation algorithm must calculate the right acquisition times for the dc-link current (depending on t_1 and t_2), and it must associate dc-link current measures with the right motor-phase currents (depending on the hexagon sector).

This estimation method is critical at low speed because of the low back-EMF and, consequently, low imposed voltage. If t_1 or t_2 is too small (less than about 10 μs), it is not possible to acquire dc-link current and then estimate motor-phase currents. It is necessary to have a time for signal stabilization and acquisition and a dead time for the inverter: the sum of these times is a minimum turn-on time value in which it is possible to measure the dc-link current.

Even at high speed, when the voltage vector moves from one sector to another, it can happen that t_1 or t_2 is not long enough. In such a case, a flag error is set up, and the current estimator will increase the corresponding current of its last incrementation value. This approach cannot be done at low speed, where several consecutive errors due to the shortness of permanence times t_1 and t_2 might occur.

Consequently, the sensorless drive, which uses only one dc-link current sensor, also requires an open-loop start-up strategy.

The estimation method proposed in this paper is suitable for both space vector pulsewidth modulation (SVPWM) [42] and DPWM and does not require any alteration in switching sequences. The authors have chosen to use DPWM, because the classic SVPWM is not performed by the platform dSPACE DS 1103, which has been used for experimental tests. In addition, DPWM has the advantage of combining the last switching state of the SVM period with the first switching state of the next SVM period; therefore, it is possible to acquire the current in the total time, which is the sum of two times "t_1" that belong to two different consecutive SMV periods.

Another estimation method is proposed in [32], where the measurement vector insertion method (MVIM) is introduced for low values of modulation index. The MVIM consists of adding three different switching states $\mathrm{V_{Ma}} = (1,0,0)$, $\mathrm{V_{Mb}} = (0,1,0)$, and $\mathrm{V_{Mc}} = (0,0,1)$, in which all three motor-phase currents can be measured, as shown in Figs. 4 and 5. However, this method increases the switching frequency of the power semiconductor devices.

To avoid problems with regard to the minimum time needed to acquire the dc-link current, the authors in [33] propose

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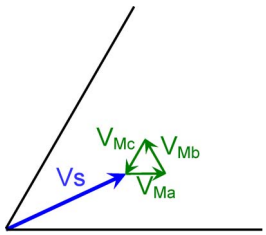


Fig. 4. MVIM description [32].

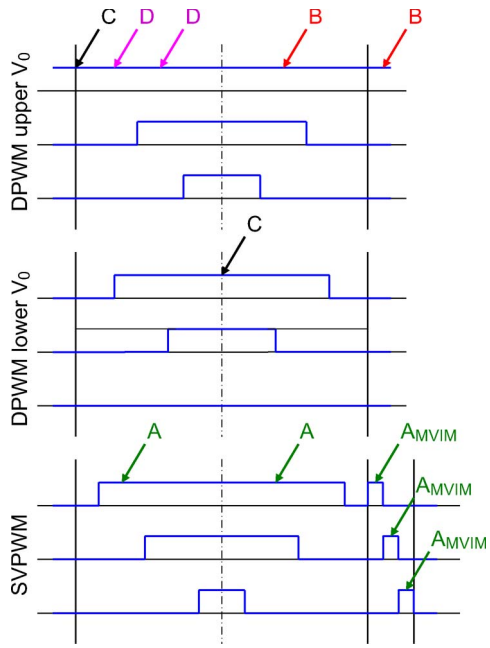


Fig. 5. DC-link current acquisitions using different current estimation methods. A is proposed in [32], B is proposed in [33], C is proposed in [34], and D is proposed in this paper.

to limit the minimum value of t_1 and t_2 to the necessary measurement time, resulting in a motor voltage distortion.

Conversely, the method used in [34] alternates the following two kinds of DPWM: 1) a DPWM with the zero voltage vector at the beginning and at the end of the SVM period and 2) a DPWM with the zero voltage vector in the middle. The dc-link current is measured at the beginning and at the center of the SVM period. The acquisition timing for each aforementioned method is shown in Fig. 5.

III. Control Strategy

This paper presents a FOC aimed at regulating the motor speed. In Fig. 6, a FOC that relies on the presence of speed/angle transducer and motor currents sensors is presented. The speed reference is compared with the measured speed, and the respective proportional–integral (PI) regulator provides the q-axis current reference; d-axis current is imposed to zero, because flux weakening operations have not been considered in this paper. The PI regulators for the d and q axis currents calculate the reference voltages that must be transformed in the α–β stationary reference frame and must be imposed through the SVM.

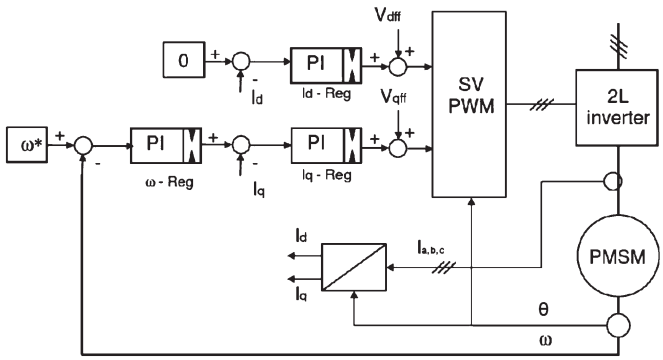


Fig. 6. Classic FOC (sensored).

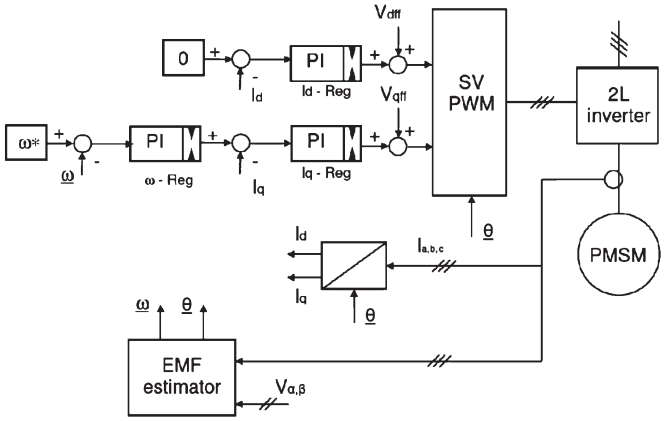


Fig. 7. Sensorless FOC.

Suitable feedforward components V_{dff} and V_{qff} are added to the output of current regulators to improve the dynamic response and limit the effects of current distortions due to the inverter action: they are calculated using the model of the PMSM in a reference frame fixed in the rotor [44]

$$\begin{cases} v_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \\ v_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e L_d i_d + K_\varphi \omega_e \end{cases} \tag{13}$$

$$\begin{cases} v_{dff} = -\omega_e L_q i_q \\ v_{qff} = \omega_e L_d i_d + K_\varphi \omega_e \end{cases} \tag{14}$$

where K_φ is a constant that depends on the motor permanent magnets.

By using the EMF estimator, the speed/angle transducer has been removed, and the new FOC scheme is presented in Fig. 7. The voltages V_α and V_β used by the EMF estimator (Figs. 7 and 8) are the α–β reference frame voltages that are originated by the PI controllers and then imposed by SVM (the intrinsic errors related to dead times, voltage drops, and delays of the inverter have been neglected).

Last, sensorless FOC with reduced current sensors is shown in Fig. 8. The only sensor in the system is a Hall-effect current transducer placed on the dc-link, between the dc-link capacitor and the inverter.

Due to this single measurement, it is possible to estimate motor-phase currents and electric angle.

It is very important to define the right sequence of control events to ensure control system stability. As shown in Fig. 9,

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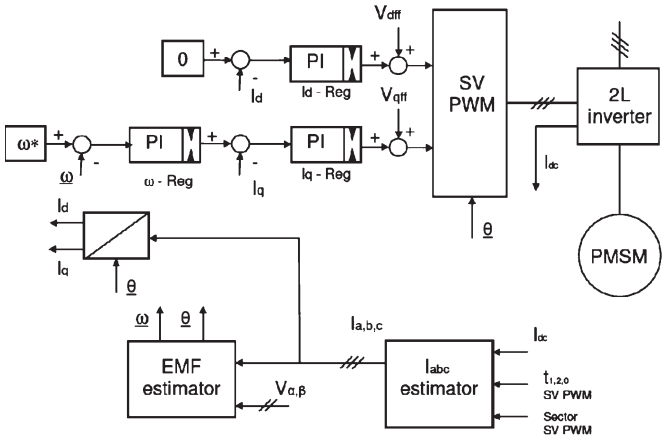


Fig. 8. Sensorless FOC with reduced current sensors.

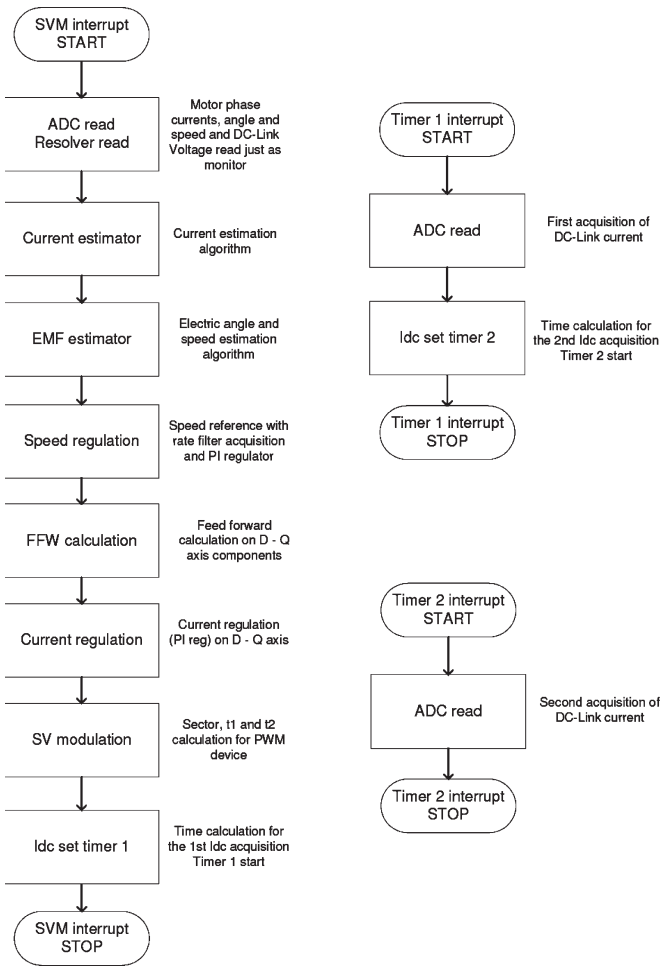


Fig. 9. Control system flowchart.

control is synchronized with the SVM interrupt that is in the middle of the commutation period. First, the current estimator is executed, followed by the EMF estimator. Then, speed regulation, feedforward calculation, and current regulation, in this sequence, precede the SVM. Finally, dc-link current acquisition times are calculated.

The procedural sequence follows the logic flux of data that are required by each control block.

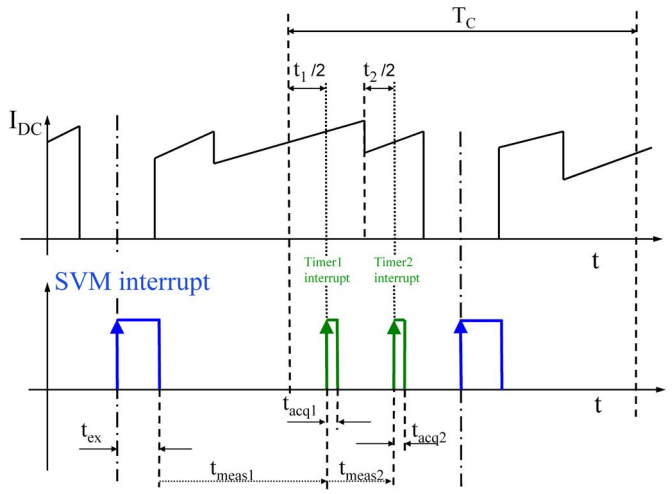


Fig. 10. DC-link current acquisition.

The two acquisitions will take place during the next SVM period; in fact, their timers are set up according to the SVM parameters calculated, which will be imposed by the inverter in the next SVM period, as shown in Fig. 10. Therefore, when SVM interrupt is generated, the current estimator associates the two dc-link current values acquired to the right motor-phase currents.

Both acquisition times t_{meas1} and t_{meas2} are calculated as a function of the execution time of control tasks t_{ex}, the SVM times t_1 and t_2, and the duration of the dc-link current sample time t_{acq1} as follows:

$$\begin{cases} t_{\mathrm{meas1}} = \frac{T_C}{2} - t_{\mathrm{ex}} + \frac{t_1}{2} \\ t_{\mathrm{meas2}} = \frac{t_1}{2} + \frac{t_2}{2} - t_{\mathrm{acq1}}. \end{cases} \tag{15}$$

IV. Starting at Low Speed

Both currents and EMF estimates are not available at low speed. Therefore, it is necessary to manage the control system in an open loop. This approach is done by the following two imposed references: 1) electrical pulsation (and angle) and 2) q-axis current (for the sensorless drive) or voltage (for the sensorless drive with reduced current sensors).

Electrical pulsation corresponds to the required speed input, but it is also limited by a rate limiter filter, because open-loop start-up, of course, cannot run at high acceleration. The electric angle is calculated by speed integration and corresponds to the phase of the imposed vector. In conventional sensorless schemes, it is possible to impose current reference and calculate voltage reference by current regulation (because motor-phase current measurements are available); the current loop is executed, but the speed loop is not.

Conversely, in single-sensor drives, it is necessary to impose voltage reference, because current estimation is not available; therefore, both current and speed loops are not executed.

A. Position Sensorless Drive

Open-loop control is operated by imposing a constant current Iq' utilizing the electric angle θ' (calculated as previously

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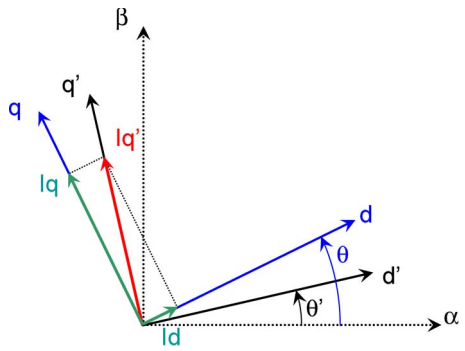


Fig. 11. Motor currents on an actual synchronous reference frame (d–q) and an imposed reference frame (d'–q').

described) in the Park transformations. In Fig. 11, two rotating reference frames are represented: the d–q reference frame variables are drawn using the real electric angle of the motor θ, which, however, is not known, whereas the $d' - q'$ reference frame is representative of the imposed angle θ'. The two reference frames rotate with the same speed, i.e., the speed imposed by the open-loop start-up strategy, but there is a difference between the angles θ and θ'. This error is due to the mismatch between the current that is needed to satisfy the actual torque request Iq and the imposed current Iq'. The motor can rotate only if the imposed current is higher than the current requested by the load torque, i.e., if $Iq' > Iq$. The excess of the imposed current, i.e., the difference between Iq' and Iq, involves an advance shift of the rotor position, and consequently, a d-axis current Id that is different from zero arises, as shown in Fig. 11. This excess of current ensures control stability, but it may cause additional power losses. Therefore, finding the best tradeoff is recommended. Conversely, whenever the imposed current Iq' is lower than Iq, a "step loss" would occur (or the motor would fail to start).

Due to these reasons, it would be better to know the load torque profile. If this information would not be available, a high value of Iq' must be chosen.

When the speed reference reaches a threshold value (about 0.15 pu), the transition from open-loop control to closed-loop control begins. The amplitude of the current reference starts to decrease, and the electric angle approaches to the estimated one. When the difference between the reference and the estimated angle is relatively low, it is possible to switch into closed-loop control, with the EMF estimator providing all necessary feedback. When transition from the open loop to the closed loop occurs, the integral part of the speed regulator is forced to assume the last value of Iq' imposed by the current regulator.

This is a good way of having a smooth transition and avoiding temporary decay on the speed profile. A suitable threshold value has been determined by simulations and experimental tests and corresponds to the minimum speed at which the sensorless algorithm ensures stability.

B. Position Sensorless Drive With Reduced Current Sensors

In this case, current measurements are not available; therefore, voltage Vq' and electric angle are directly imposed to the SVM. Its value is proportional to the speed reference, i.e.,

$$\left|V_q'\right| = k_1 + k_2 \cdot \omega_{\text{ref}}. \tag{16}$$

The parameters k_1 and k_2 have been set to have a current profile similar to the position sensorless case.

The k_1 term is necessary to compensate for resistive voltage drop, which is predominant at low frequency. This kind of start-up strategy is similar to the volts per hertz strategy. Transitions between open and closed loops are made similar to the "position sensorless" case at about 0.4 pu rotor speed.

TABLE I
MOTOR DATA

Symbol	Quantity	Value
P_n	Nominal power	1 kW
V	Nominal source voltage (RMS)	200 V
T	Nominal torque	4 N m
ω	Nominal speed	3000 rpm
$2p$	Numbers of poles	8
J	Moment of inertia	3.5 * 10^{-4} kg·m^2
Φ	Nominal flux linkage	0.67 V·s
R_s	Stator resistance at T = 20°C	0.65 Ω
L_s	Stator inductance	2.4 mH

V. SIMULATION AND EXPERIMENTAL RESULTS

This method has been investigated through simulation prior to the hardware implementation of the system. The MATLAB/Simulink environment has been used for the simulation, linked to the Piecewise Linear Electrical Circuit Simulation (PLECS) circuital simulator.

The parameter values used in the simulation scope were equal to the motor model and control configuration.

Control and estimation algorithms have been implemented in the American National Standards Institute (ANSI) C code (through S-Function) so that the same code used in the simulation is fully portable to the experimental system. This approach allows an excellent closeness between the simulation and the real system.

The main data of the synchronous motor used in this paper are reported in Table I. An insulated-gate bipolar transistor (IGBT) inverter with a nominal power of 10 kVA, switching frequency of the power semiconductor devices equal to 5 kHz, and dc-link nominal voltage of 350 V has been used. The control strategy has been implemented on dSPACE DS 1103 with (333-MHz) PowerPC; this processor can directly be programmed in ANSI C code, thereby avoiding the MATLAB real-time workshop automatic code generation. This approach leads to a greater efficiency in code execution. The execution time is lower than 40 μs, and it also includes time for the measurements of motor currents and speed/position, which have been done only to compare the estimated variables with the measured variables. If these extra measurements are removed, the execution time of estimators and control algorithm is about 13 μs.

The acquisition system is based on the Control Desk interface, and the data elaboration is performed in a MATLAB environment.

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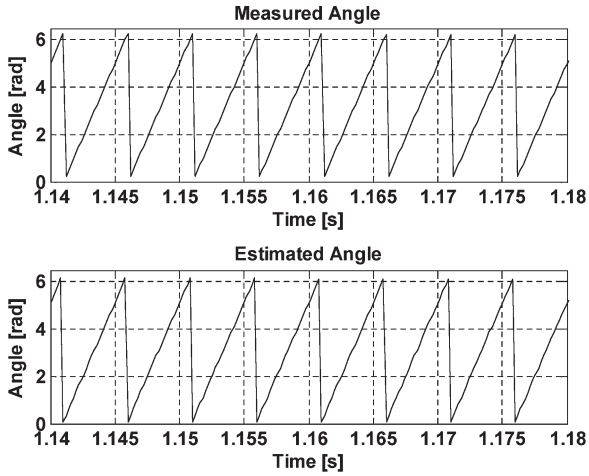


Fig. 12. Measured and estimated angles in the simulation results at nominal speed.

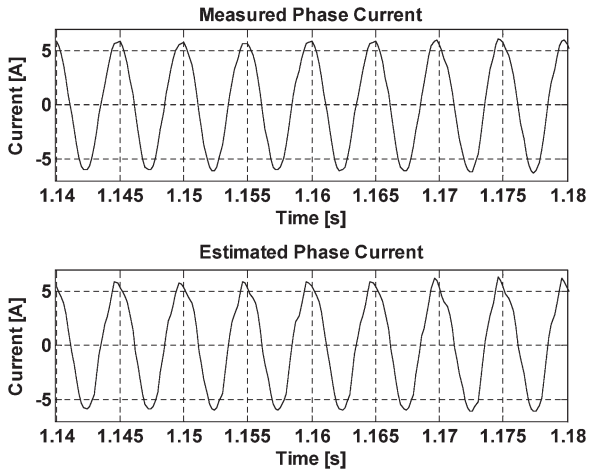


Fig. 13. Measured and estimated currents in the simulation results at nominal speed.

The main control task is started by SVM interrupt, and it runs at the switching frequency of 5 kHz.

The machine used as a brake is a dc generator with a nominal power of 1.1 kW and a nominal current of 5 A. It emulates a viscous load with the torque proportional to the speed.

A resolver is connected to the motor shaft, but it has been used only to test the position and speed estimates or to compare the performance of the position sensorless drive or the position sensorless drive with reduced current sensors with the full sensored drive performance.

A. Simulation Results

Simulation in the sensorless drive with reduced current sensors configuration exhibits an excellent response, and Fig. 12 shows how, at nominal speed, the angle estimate follows the real rotor position. The current estimator quite faithfully reproduces the phase motor current, as shown in Fig. 13.

The slight distortion of the estimated current, as reported in Fig. 13, lower trace, is related to the dc-link current acquisition times t_{meas1} and t_{meas2}, shown in Fig. 10. Such times depend on the space vector permanence times t_1 and t_2, and then, the current acquisition is not always made in the same instants within the SVM period. This approach creates the estimated current distortion. In addition, experimental results confirm this distortion.

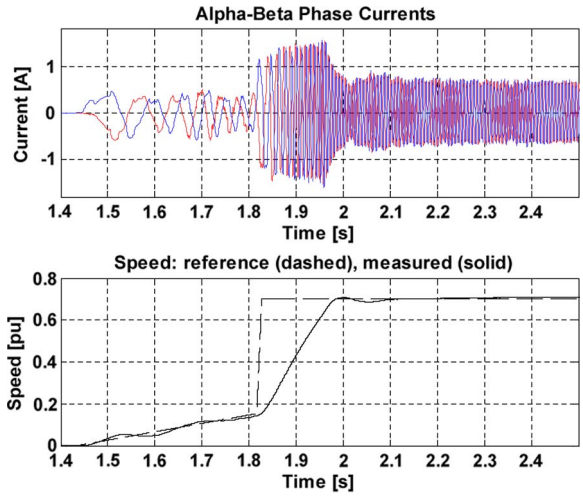


Fig. 14. Position sensorless: Start-up from 0 pu to 0.7 pu.

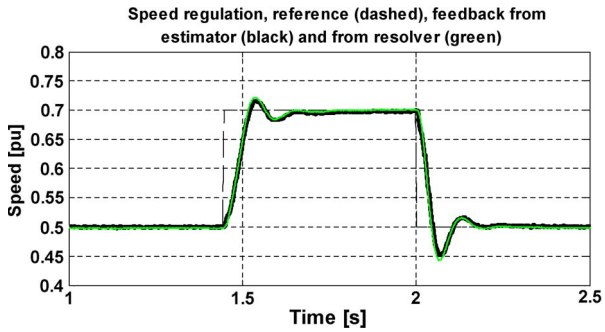


Fig. 15. Position sensorless: Dynamic response with speed reference variations.

Simulations with motor parameters ($\mathrm{R_s}$ and $\mathrm{L_s}$) variation have been performed, but the current estimation is not affected in an appreciable way.

B. Experimental Results

First, results related to the position sensorless drive are presented. Fig. 14 shows a speed reference variation from 0 pu to 0.7 pu, where two different values of the maximum slope, imposed by the rate limiter filter in the open and closed loops, can be observed.

When the control strategy goes into the closed loop, dynamic response becomes comparable to the sensored drive one. In Fig. 15, the motor dynamic performance related to speed reference step variations is presented.

The control system ensures a prompt speed response, and it can be noted how the estimated speed is very close to the measured one.

Second, a few results related to the sensorless drive with reduced current sensors are presented.

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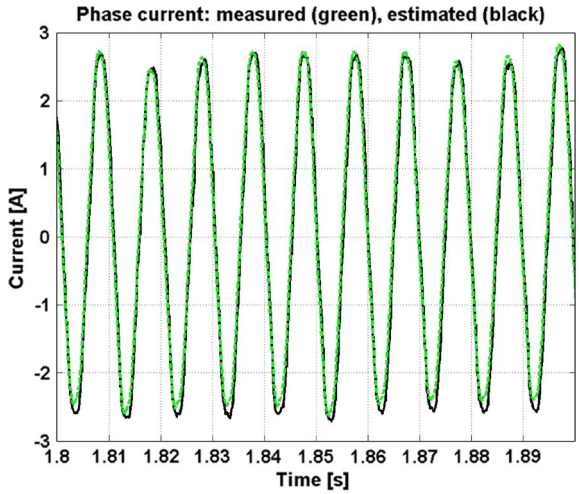


Fig. 16. Measured and estimated current at 0.5 pu speed.

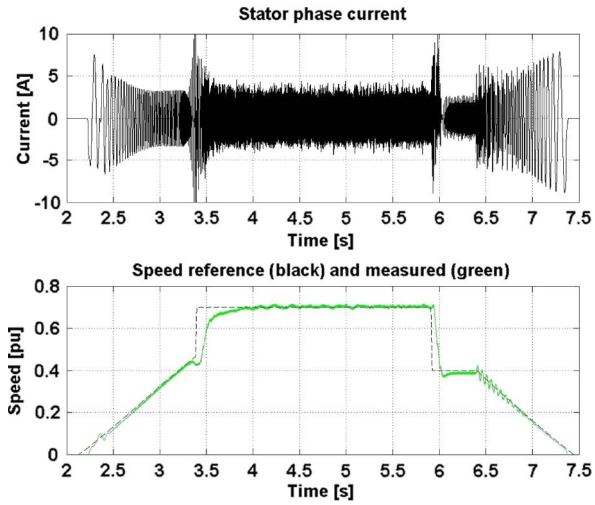


Fig. 17. Reduced current sensors drive: Start-up from 0 pu to 0.7 pu and stop.

Fig. 16 shows the good accuracy of current estimation at a constant speed, equal to 0.5 pu. Start-up from 0 pu to 0.7 pu and subsequent return to zero speed is presented in Fig. 17. In this configuration, the open-loop speed range has been designed to be wider than the speed range related to the position sensorless drive because of the replacement of current sensors with the current estimator; however, the requested speed is promptly reached.

When the motor runs in closed-loop operations, it is possible to obtain good dynamic performance, absolutely comparable to the performance related to a full sensored drive. Fig. 18 shows motor speed behavior in the presence of reference step variations, from 0.5 pu to 0.7 pu, and vice versa. To better compare dynamic performance among sensored (C), position sensorless (B), and sensorless with reduced current sensors (A) drives, the three speed or current traces, related to a speed reference step variation from 0.5 pu to 0.7 pu, are reported as follows. This speed range ensures the activation of closed-loop control strategy in both sensorless drives.

In Fig. 19, a comparison among speed traces is presented, where the same values of PI regulator gains are used. These

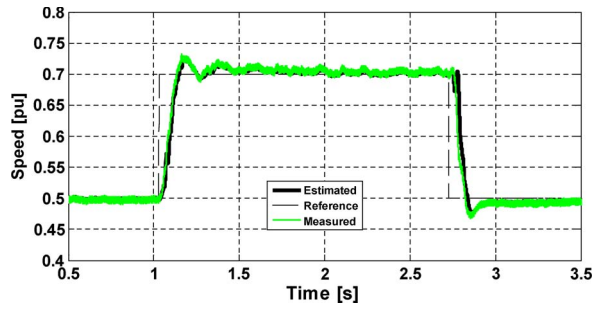


Fig. 18. Reduced-current-sensors drive: Dynamic response with speed reference variations.

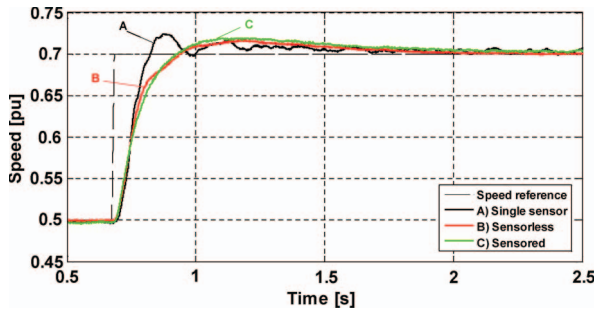


Fig. 19. Comparison among the sensorless drive with reduced current sensors (A), position sensorless drive (B), and sensored (C) drives using the same PI regulator parameters in case of a speed reference step variation from 0.5 pu to 0.7 pu.

TABLE II
SPEED AND CURRENT REGULATOR PARAMETER RELATIVE VALUES

FOC strategy	Speed regulator		Current regulator	
	Ki	Ki / Kp	Ki	Ki / Kp
A) Single sensor	$k_{iw,A}$	$\frac{k_{iw,A}}{k_{pw,A}}$	$k_{ii,A}$	$\frac{k_{ii,A}}{k_{pi,A}}$
B) Sensorless	$4\cdot k_{iw,A}$	$1\cdot\frac{k_{iw,A}}{k_{pw,A}}$	$1\cdot k_{ii,A}$	$0.5\cdot\frac{k_{ii,A}}{k_{pi,A}}$
C) Sensored	$5\cdot k_{iw,A}$	$1\cdot\frac{k_{iw,A}}{k_{pw,A}}$	$1\cdot k_{ii,A}$	$0.5\cdot\frac{k_{ii,A}}{k_{pi,A}}$

values have been optimized with reference to the more critical drive, which is the sensorless drive with reduced current sensors, to have stability for all drives. The trace related to the proposed single sensor method exhibits slight oscillations at steady-state speed, which are consequence of current estimation distortions.

Conversely, if parameters are set to obtain the best performance from each drive, as shown in Table II, which gives the relative values of the controller parameters among each others, with reference to the generic PI regulator transfer function reported in (17), dynamic responses of the position sensorless and sensored drives improve, as shown in Fig. 20. PI regulator parameters have been calculated using classical stability criteria, also validated with the simulator, and then adjusted during

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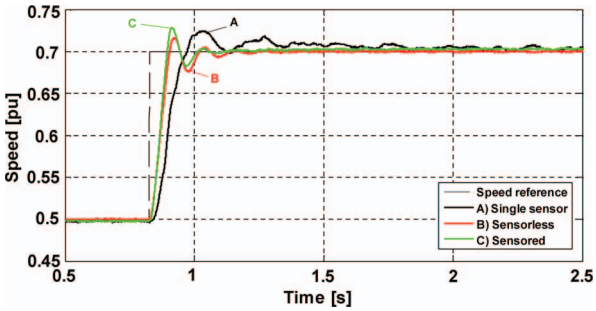


Fig. 20. Comparison among sensorless with reduced current sensors (A), position sensorless (B), and sensored (C) drives, each with its best PI regulator parameters, in case of a speed reference step variation from 0.5 pu to 0.7 pu.

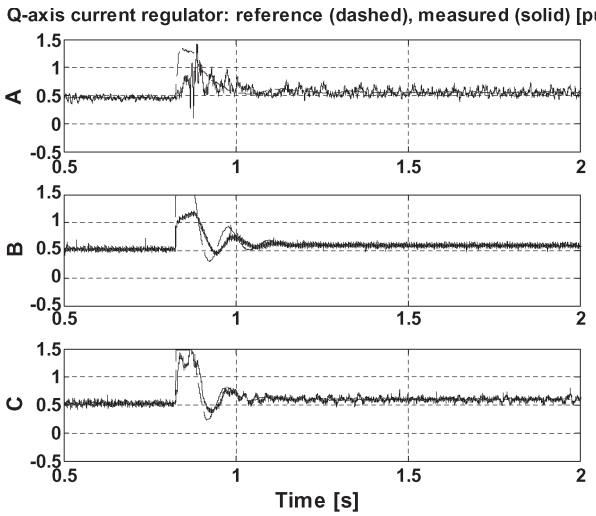


Fig. 21. q-axis current in sensorless with reduced current sensors (A), position sensorless (B), and sensored (C) drives, each with its best PI regulator parameters, in case of a speed reference step variation from 0.5 pu to 0.7 pu.

the experimental tests to obtain the best tradeoff between a good stability margin and high dynamic performance.

$$T_{PI_reg(s)} = K_i \frac{1 + s\frac{K_p}{K_i}}{s}. \tag{17}$$

Note that there is a very slight difference between sensored and position sensorless speed profiles. In addition, the sensorless drive with reduced current sensors exhibits always a quick response. The q-axis current behaviors related to this last test are also reported, as shown in Fig. 21. Note that the q-axis current reference in the sensorless drive with reduced current sensors does not saturate to the maximum allowed value, whereas in both position sensorless and sensored drives, it does. This case is because the speed regulator gain in the single sensor drive is lower than in the other two drives. With regard to the q-axis current and, therefore, torque response, a slight worsening in a single sensor drive can be appreciated, even if the overall drive performance remains more than acceptable.

Finally, the dynamic response to rated load torque variations at constant speed, equal to 0.5 pu, is presented in Figs. 22 and 23. The drive is in a single sensor configuration, and the fast response of current regulators and, therefore torque, as shown in Fig. 22, confirms the robustness of the proposed algorithm.

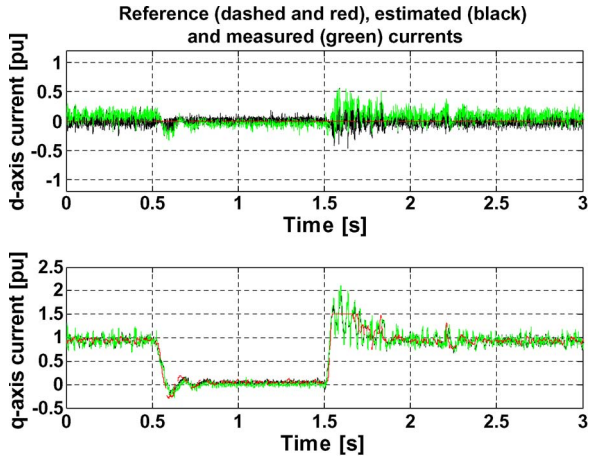


Fig. 22. Reduced-current-sensors drive: Dynamic response with rated load torque variations at constant speed (0.5 pu), d–q axis and reference, estimated, and measured currents.

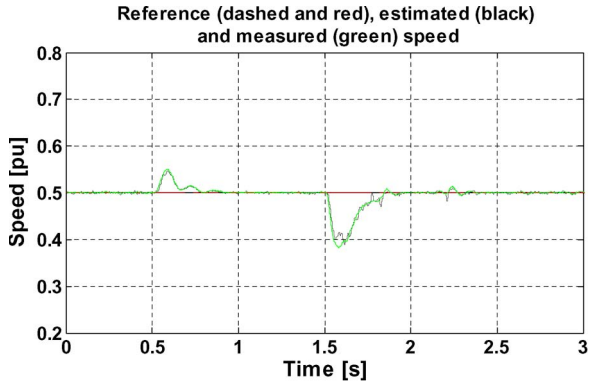


Fig. 23. Reduced-current-sensors drive: Dynamic response with rated load torque variations at constant speed (0.5 pu) and reference, estimated, and measured speed.

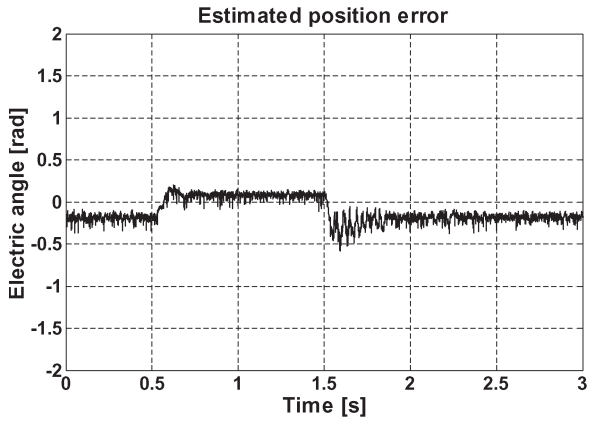


Fig. 24. Reduced-current-sensors drive: Dynamic response with rated load torque variations at constant speed (0.5 pu) and angular difference between the actual and estimated frames.

The angular difference between the actual and estimated frames during the same test conditions is reported in Fig. 24.

VI. Conclusion

Sensorless drives may exhibit high dynamic responses, as provided by sensored drives. In this paper, it has been shown

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that, using an accurate rotor speed and angle estimator, decreasing the total costs becomes possible, without appreciable deterioration of dynamic performance.

It is possible to further reduce the drive cost by reducing the number of current sensors, introducing a motor-phase current estimator that uses only a dc-link current sensor.

A comparison among full sensored drive, position sensorless drive, and position sensorless drive with only one dc-link current sensor has been performed, and experimental results confirmed that the latter strategy also leads to a high dynamic response in the medium- to high-speed range. Sensorless drives with reduced current sensors are suitable for general-purpose and low-cost applications, where an excellent dynamic response in all speed range is not requested.

In particular, the proposed drive system seems to be appropriate to face load torque profiles typical of viscous or aerodynamical load, where the resistant torque is well defined *a priori*.

However, even in the case of a load that requires unknown torque variations, the drive can operate in all the speed range, even if with a reduced energy efficiency.

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